BME302 Unit II – Fluid Kinematics: Types of Flow & Continuity

Lecture Notes · Fluid Mechanics and Fluid Machines

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How to use these notes. Every topic, worked example and question carries one tag. CORE must-know syllabus material at AKTU exam level: master these first. HOTS higher-order thinking: analyse, compare, justify or design (Bloom K4 to K6). ADVANCED beyond the exam level for most students: shaded orange, optional reading.

Unit overview

Unit II is fluid kinematics: describing how a fluid moves, without yet asking which forces cause the motion. It sorts flows into the standard pairs of types that decide which equations apply, then derives the continuity equation, the law of conservation of mass written for a flowing fluid. It maps to **CO1: Understand the application of mass and momentum conservation laws for fluid flows (K2)** and is planned for 8 lectures.

Lecture	Topic	Key outcome
L1	Lagrangian and Eulerian descriptions; continuum and free molecular flows	Choose how to describe a flow
L2	Steady and unsteady, uniform and non-uniform flows	Classify a flow in time and space
L3	Laminar and turbulent flows; Reynolds number	Predict the flow regime
L4	Rotational and irrotational flows	Test a velocity field for rotation
L5	Compressible and incompressible; subsonic, sonic and supersonic flows	Use the Mach number
L6	Sub-critical, critical and supercritical flows; 1D, 2D and 3D flows	Use the Froude number; count dimensions
L7	Streamlines, pathlines, streaklines, stream tubes	Find the equation of a streamline
L8	Continuity equation in 3D and 1D	Apply conservation of mass

Common notation: u, v, w = velocity components in the x, y, z directions; V = velocity magnitude or mean velocity; t = time; ρ = density; A = area; Q = discharge; $\dot{m}$ = mass flow rate; Re = Reynolds number; M = Mach number; Fr = Froude number; ω = angular velocity

of rotation; ζ = vorticity; Kn = Knudsen number.

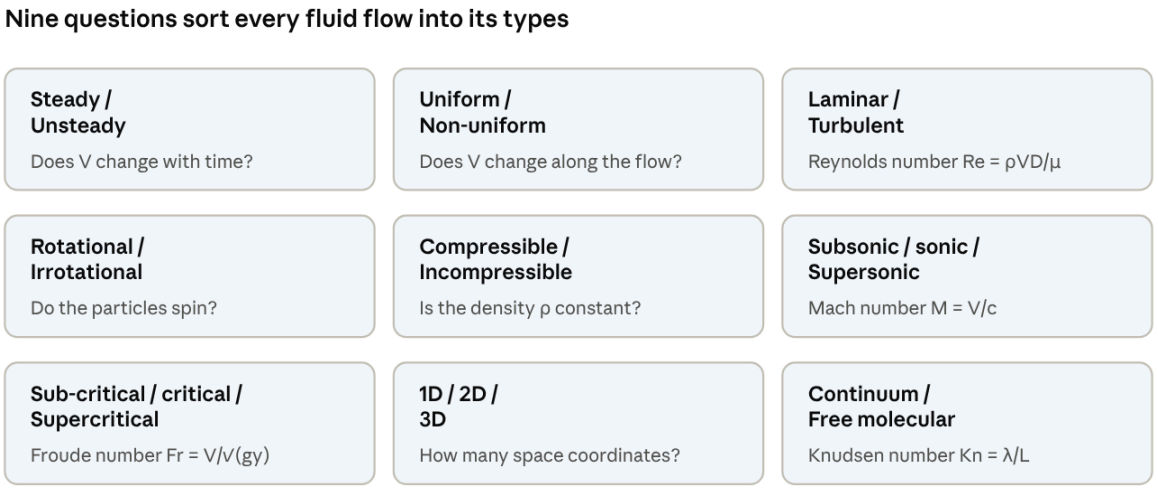


Figure 1. Types of fluid flow and the question or number that decides each

The pairs are independent: a single flow can be steady, non-uniform, turbulent, rotational, incompressible and three-dimensional all at once.

0. Before you start: basic ideas you need CORE

Unit II describes how fluid moves. Before the sections begin, this part builds the four ideas you need: what a velocity field is, what mass conservation means, how to differentiate a field, and what the dimensionless numbers tell you.

0.1 Velocity, acceleration and fields CORE

> **IN SIMPLE WORDS**
>
> A **field** is a quantity that has a value at every point in space. A weather map of temperature is a field. The velocity of a flowing fluid is also a field: at each point the fluid moves with some speed in some direction. We split that velocity into three parts. The letter u is the speed along x, v along y and w along z. Acceleration is how fast the velocity changes, in $\mathrm{m/s^2}$.

Symbol	Meaning	SI unit
x, y, z	Position coordinates	m
t	Time	s
u, v, w	Velocity components along x, y, z	m/s
V	Magnitude of velocity, $V = \sqrt{u^2 + v^2 + w^2}$	m/s
a_x, a_y, a_z	Acceleration components	$\mathrm{m/s^2}$

Example 0.1 CORE The velocity field is $u = 2x$, $v = y$ (in m/s, with x and y in m). Find the velocity at the point $(1, 2)$.

1. $u = 2 \times 1 = 2\,\mathrm{m/s}$; $v = 2\,\mathrm{m/s}$
2. $V = \sqrt{u^2 + v^2} = \sqrt{2^2 + 2^2} = 2.83\,\mathrm{m/s}$

0.2 Control volume and mass conservation CORE

IN SIMPLE WORDS

A **control volume** is an imaginary box fixed in space. Fluid flows in through some faces and out through others. Mass can neither appear nor vanish. So mass in minus mass out equals the mass that piles up inside. Think of a bath tub with the tap running and the plug open: if more water flows in than out, the level rises. The continuity equation is this bookkeeping written with symbols.

$$\text{Rate of mass in} - \text{Rate of mass out} = \text{Rate of increase of mass inside}$$

For steady flow nothing accumulates, so mass in equals mass out.

Example 0.2 CORE A tank receives water at $5\,\mathrm{kg/s}$ and loses $3\,\mathrm{kg/s}$ through an outlet. How fast does the mass in the tank increase?

1. $\text{Rate of increase} = 5 - 3 = 2\,\mathrm{kg/s}$

0.3 Partial derivatives and the total derivative CORE

IN SIMPLE WORDS

A **partial derivative** such as $\partial u/\partial x$ asks how fast u changes with x alone, while all other variables are held fixed. A fluid particle is different: it moves, so it meets new values of u as time passes and as it travels. Imagine walking along a road on a hot day. The temperature you feel changes because the day warms up (time) and because you reach a shadier spot (position). The **total derivative** adds both effects. This is how we get the acceleration of a fluid particle.

Derivation from scratch: acceleration of a fluid particle

Aim. Find the acceleration components a_x, a_y, a_z of a fluid particle in terms of the velocity field $u(x, y, z, t)$, $v(x, y, z, t)$, $w(x, y, z, t)$.

Assumptions.

1. The Eulerian velocity field is known and its derivatives exist.
2. We follow one fluid particle; in a small time dt it moves with the local velocity.

Step 1. Position change of the particle. In time dt the particle moves a distance equal to velocity times time in each direction.

$$dx = u\,dt, \qquad dy = v\,dt, \qquad dz = w\,dt$$

Step 2. Change of u seen by the particle. The component u depends on four variables. The change of a function of several variables is the sum of the changes caused by each variable (the chain rule).

$$du = \frac{\partial u}{\partial t}\,dt + \frac{\partial u}{\partial x}\,dx + \frac{\partial u}{\partial y}\,dy + \frac{\partial u}{\partial z}\,dz$$

Step 3. Substitute the position changes. Put $dx = u\,dt$, $dy = v\,dt$ and $dz = w\,dt$ from Step 1.

$$du = \frac{\partial u}{\partial t}\,dt + \frac{\partial u}{\partial x}\,u\,dt + \frac{\partial u}{\partial y}\,v\,dt + \frac{\partial u}{\partial z}\,w\,dt$$

Step 4. Divide by dt. The acceleration is the rate of change of velocity, $a_x = du/dt$.

$$a_x = \frac{du}{dt} = \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z}$$

Step 5. Repeat for v and w. The same argument for v and w gives the other two components.

Result.

$$a_x = \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z}$$

$$a_y = \frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z}$$

$$a_z = \frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z}$$

What it tells us. The first term is the **local acceleration**: velocity changing with time at a fixed point. The other three terms are the **convective acceleration**: the particle moves to a place where the velocity is different. A steady flow has zero local acceleration, but a particle in a tapering pipe still accelerates.

Example 0.3 CORE The one-dimensional flow has $u = 2x + t$ (SI units) and $v = w = 0$. Find the acceleration a_x at $x = 1\ \mathrm{m}$, $t = 1\ \mathrm{s}$.

1. $\dfrac{\partial u}{\partial t} = 1$; $\dfrac{\partial u}{\partial x} = 2$; $u = 2\,(\,1\,) + 1 = 3\ \mathrm{m/s}$
2. $a_x = \dfrac{\partial u}{\partial t} + u\dfrac{\partial u}{\partial x} = 1 + 3 \times 2 = 7\ \mathrm{m/s^2}$

0.4 Units and typical values CORE

IN SIMPLE WORDS

Three dimensionless numbers appear in this unit. The Reynolds number Re compares inertia with viscous force. The Mach number M compares flow speed with the speed of sound. The Froude number Fr compares flow speed with the speed of a surface wave. They have no units, so the same value means the same kind of flow in any unit system. Always put every quantity in SI units before using them.

Number	Formula	Compares	Typical value
Re	VD/ν	Inertia to viscous force	Water in a pipe: 10^4 to 10^6
M	V/c	Flow speed to speed of sound	Passenger jet at cruise: about 0.8 (air at 15 °C has $c \approx 340\ \mathrm{m/s}$)
Fr	$V/\sqrt{gy}$	Flow speed to surface-wave speed	Slow river: below 1

Convert	To	Rule	Example
mm	m	Divide by 1000	$50\ \mathrm{mm} = 0.05\ \mathrm{m}$
km/h	m/s	Divide by 3.6	$120\ \mathrm{km/h} = 33.3\ \mathrm{m/s}$
stokes	$\mathrm{m^2/s}$	Multiply by 10^{-4}	$0.5\ \mathrm{St} = 5 \times 10^{-5}\ \mathrm{m^2/s}$
°C	K	Add 273	$15\ ^\circ\mathrm{C} = 288\ \mathrm{K}$

Example 0.4 CORE Water ($\nu = 10^{-6}\ \mathrm{m^2/s}$) flows at $1\ \mathrm{m/s}$ in a 50 mm pipe. Find Re.

1. $\mathrm{Re} = \dfrac{VD}{\nu} = \dfrac{1 \times 0.05}{10^{-6}} = 50\,000$: far above 4000, so **turbulent**

Example 0.5 CORE Air flows at $100\ \mathrm{m/s}$ where $c = 340\ \mathrm{m/s}$. Find M and say if the flow can be treated as incompressible.

1. $M = \dfrac{100}{340} = 0.294$, below 0.3: **incompressible** treatment is acceptable

Check your understanding.

1. CORE What does $\partial u/\partial x$ mean? *(How fast u changes with x alone, other variables held fixed.)*
2. CORE A steady flow has zero local acceleration. Can a particle still accelerate? *(Yes, through the convective terms, e.g. in a tapering pipe.)*
3. CORE A 120 km/h wind speed is how many m/s? *(Divide by 3.6: $33.3\ \mathrm{m/s}$.)*
4. HOTS Why must data be converted to SI before computing Re? *(Re is only unit-free if all quantities use consistent units.)*

1. Describing fluid motion; continuum and free molecular flows (L1) CORE

Fluid motion is described either by following individual particles or by watching fixed points in space. Fluid mechanics almost always uses the second, the Eulerian description, which gives the velocity field $V(\,x, y, z, t\,)$.

IN SIMPLE WORDS

There are two ways to watch a river. You can float along with a leaf, or you can stand on a bridge and watch the water pass one spot. The second way is the Eulerian view, and it is the one we use. It gives the velocity at each fixed point. Fluid mechanics also needs the fluid to be smooth, not a swarm of molecules. That smooth picture is the continuum.

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

1.1 Lagrangian and Eulerian descriptions ADVANCED

Description	What is followed	Variables	Example
Lagrangian	One fluid particle as it moves	Position of the particle as a function of time	A float released in a river and tracked
Eulerian	Fixed points in space as fluid passes	Velocity, pressure, density as functions of (x, y, z, t)	A flow meter fixed in a pipe

In the Eulerian description the velocity field is written as:

$$\vec{V} = u(x, y, z, t)\,\hat{i} + v(x, y, z, t)\,\hat{j} + w(x, y, z, t)\,\hat{k}$$

1.2 Continuum and free molecular flows CORE

IN SIMPLE WORDS

Air is made of molecules with empty space between them. If the gaps are tiny compared with the object, the air feels like a smooth material, a continuum. At very high altitude the gaps are huge and molecules hit the body one at a time. The Knudsen number Kn is the ratio of the gap size (λ, the mean free path) to the size of the body (L). A small Kn means continuum.

A fluid is made of molecules separated by gaps. When the gaps are tiny compared with the size of the body or channel, the fluid behaves as a **continuum**: density and velocity can be defined at a point and vary smoothly. When the gas is so rarefied that molecules seldom collide with each other, they act individually on the body: **free molecular flow**. The deciding ratio is the Knudsen number, the mean free path λ over a characteristic length L:

$$\mathrm{Kn} = \frac{\lambda}{L}$$

Kn	Regime	Treatment	Example
< 0.01	Continuum flow	Navier–Stokes equations, no slip at walls	Almost all engineering flows
0.01–0.1	Slip flow	Continuum, with velocity slip at walls	Micro-channels, MEMS devices
0.1–10	Transition flow	Kinetic theory (Boltzmann, DSMC)	Re-entry vehicles at high altitude
> 10	Free molecular flow	Individual molecule–wall collisions	Satellites in low Earth orbit, vacuum pumps

For air at sea level $\lambda \approx 68\ \text{nm}$, so for a 1 mm channel $\text{Kn} \approx 7 \times 10^{-5}$: firmly a continuum. Everything else in this course assumes a continuum.

Common mistakes and exam tips.

- Do not call the Eulerian description "following a particle". Eulerian means fixed points; Lagrangian means following a particle.
- $\text{Kn} = \lambda / L$ is a ratio of two lengths. Use the same unit for both (nm and mm must be converted to the same unit).
- A small Kn (below 0.01) means continuum. Students often reverse this.
- Remember the numbers: continuum below 0.01, free molecular above 10.

Check your understanding.

1. CORE Which description fixes the observer at a point in space? *(The Eulerian description.)*
2. CORE Is $\text{Kn} = 10^{-4}$ a continuum flow? *(Yes, since it is below 0.01.)*
3. HOTS Why do satellites in low Earth orbit need free molecular theory? *(The air is so thin that λ is large compared with the satellite, so $\text{Kn} > 10$.)*

2. Steady and unsteady, uniform and non-uniform flows (L2)

CORE

These two pairs ask different questions. Steady or unsteady: does the flow at a fixed point change with **time**? Uniform or non-uniform: does the velocity change from point to point in **space**, at a given instant?

IN SIMPLE WORDS

Two separate questions are asked about any flow. First: is it changing with time? Second: is it changing from place to place? A tap running at a constant rate is steady. A tap being opened slowly is unsteady. A garden hose of constant width is uniform. A hose that narrows at the nozzle is non-uniform, because the water speeds up along it.

2.1 Steady and unsteady flow CORE

> **IN SIMPLE WORDS**
> Stand at one spot and watch the water go by. If the speed, pressure and density you see never change, the flow is steady. If they change as the minutes pass, it is unsteady. The partial derivative $\partial V/\partial t$ means "change with time at a fixed point". Steady flow means this is zero.

In **steady flow** the velocity, pressure and density at a point do not change with time; in **unsteady flow** they do:

$$\text{Steady: } \left(\frac{\partial V}{\partial t}\right)_{x_0, y_0, z_0} = 0,\ \left(\frac{\partial p}{\partial t}\right)_{x_0, y_0, z_0} = 0,\ \left(\frac{\partial \rho}{\partial t}\right)_{x_0, y_0, z_0} = 0 \qquad \text{Unsteady: } \frac{\partial V}{\partial t} \neq 0$$

Turbulent flow, whose velocity fluctuates rapidly at every point, is treated as steady when its time-averaged values are constant.

2.2 Uniform and non-uniform flow CORE

> **IN SIMPLE WORDS**
> Now freeze time, like a photograph, and compare different points along the flow. If the velocity is the same everywhere, the flow is uniform. If it differs, the flow is non-uniform. The derivative $\partial V/\partial s$ means "change of speed with distance s along the flow at one instant". Uniform flow means this is zero.

In **uniform flow** the velocity is the same in magnitude and direction at every point along the flow at a given instant; in **non-uniform flow** it changes with position s:

$$\text{Uniform: } \left(\frac{\partial V}{\partial s}\right)_{t\,=\,\text{const}} = 0 \qquad \text{Non-uniform: } \left(\frac{\partial V}{\partial s}\right)_{t\,=\,\text{const}} \neq 0$$

2.3 The four combinations HOTS

Combination	Meaning	Example
Steady uniform	Same velocity everywhere, at all times	Constant discharge through a long straight pipe of constant diameter
Steady non-uniform	Velocity varies along the flow, but not with time	Constant discharge through a tapering pipe or a venturimeter
Unsteady uniform	Same velocity everywhere, changing with time	A constant-diameter pipe while a pump speeds up
Unsteady non-uniform	Varies with position and time	Flow in a tapering pipe while a valve is being closed; flood waves

Common mistakes and exam tips.

- Steady is about **time**; uniform is about **space**. Do not mix them up.
- A tapering pipe with constant discharge is steady and **non-uniform**. This is the most common exam example.
- In steady flow the velocity at a point does not change with time, but a particle can still accelerate by moving to a faster region.
- For each of the four combinations, learn one example (the table above).

Check your understanding.

1. CORE In which type of flow is $\partial V/\partial t = 0$? *(Steady flow.)*
2. CORE Is constant flow through a venturimeter uniform? *(No. The area changes, so the velocity changes with position: steady non-uniform.)*
3. HOTS Can a flow be unsteady yet uniform? *(Yes: a constant-diameter pipe while a pump speeds up.)*

3. Laminar and turbulent flows (L3) CORE

In **laminar flow** the fluid moves in smooth layers (laminae) that slide over one another without mixing. In **turbulent flow** the particles move in irregular, eddying paths and the layers mix vigorously. Which one occurs is decided by the Reynolds number, the ratio of inertia forces to viscous forces.

IN SIMPLE WORDS

Smoke from an incense stick rises in a thin straight line, then breaks into swirls higher up. The straight part is laminar flow. The swirling part is turbulent flow. Slow, thick, sticky fluids stay laminar. Fast, thin fluids turn turbulent. The Reynolds number Re is one number that tells you which one to expect.

3.1 Reynolds' experiment (1883) CORE

IN SIMPLE WORDS

Reynolds made the flow visible. He put a thin thread of coloured dye into water running through a glass tube. At low speed the thread stayed a straight line. At higher speed it wobbled. At high speed it smeared across the whole tube. So the same tube can show all three types, depending only on the speed.

Water flows from a tank through a glass tube with a bell-mouth entrance; a thin filament of dye is injected at the inlet and the flow is controlled by a valve at the outlet.

1. **Low velocity:** the dye forms a straight, steady line along the tube: laminar flow.
2. **Intermediate velocity:** the dye line begins to wave and break up intermittently: transition.
3. **High velocity:** the dye spreads across the whole tube soon after entry: turbulent flow.

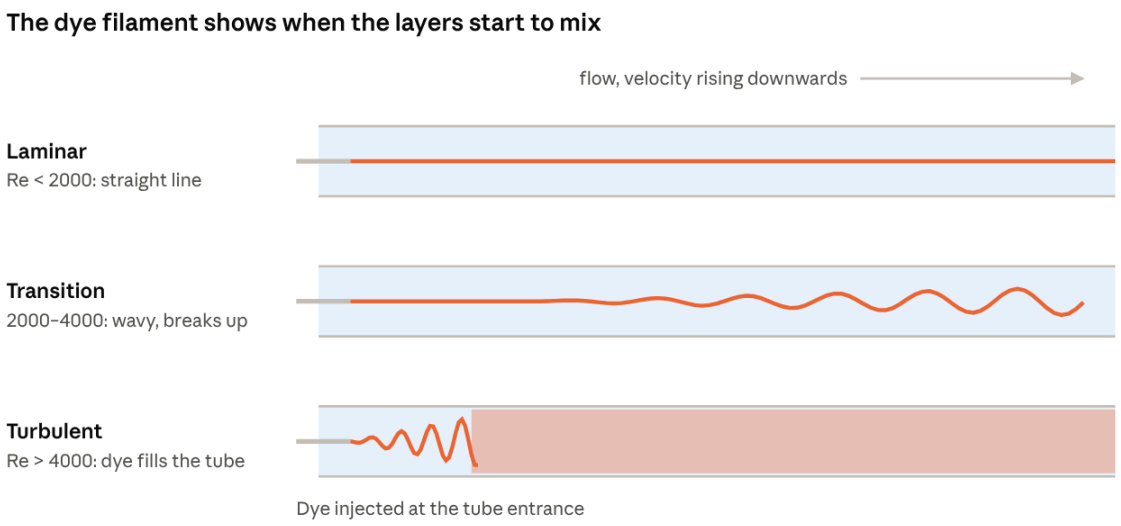


Figure 2. Reynolds' experiment: the dye filament in laminar, transition and turbulent flow

3.2 Reynolds number CORE

IN SIMPLE WORDS

Re compares two things: how strongly the moving fluid "wants to keep going" (inertia) and how strongly its stickiness resists (viscosity). In $\mathrm{Re} = VD/\nu$, V is the speed, D is the pipe diameter and ν is the kinematic viscosity. A big Re means inertia wins, so the flow is turbulent. A small Re means viscosity wins, so the flow is smooth.

$$\mathrm{Re} = \frac{\text{inertia force}}{\text{viscous force}} = \frac{\rho V D}{\mu} = \frac{VD}{\nu}$$

Derivation from scratch: Reynolds number as a ratio of forces

Aim. Show that the ratio of inertia force to viscous force is $\rho VD/\mu$.

Assumptions.

1. Take a small blob of fluid whose size is a characteristic length L (for a pipe, $L = D$).
2. The speed is V over the length L, so the velocity gradient is of order V/L.

Step 1. Inertia force. Inertia force is mass times acceleration. The mass is density times volume. The acceleration is the change in speed V over the time L/V the blob needs to move its own length.

$$m \propto \rho L^3, \qquad a \propto \frac{V}{L/V} = \frac{V^2}{L}$$

$$F_{\text{inertia}} = m\,a \propto \rho L^3 \times \frac{V^2}{L} = \rho V^2 L^2$$

Step 2. Viscous force. By Newton's law of viscosity the shear stress is $\tau = \mu\, du/dy$, which is of order $\mu V/L$. It acts on an area of order L^2.

$$F_{\text{viscous}} = \tau \times \text{Area} \propto \mu \frac{V}{L} \times L^2 = \mu V L$$

Step 3. Take the ratio. Divide Step 1 by Step 2 and cancel V and L.

$$\frac{F_{\text{inertia}}}{F_{\text{viscous}}} = \frac{\rho V^2 L^2}{\mu V L} = \frac{\rho V L}{\mu}$$

Step 4. Put $L = D$ and use $\nu = \mu/\rho$.

$$\frac{\rho V D}{\mu} = \frac{V D}{\nu}$$

Result.

$$\text{Re} = \frac{\text{inertia force}}{\text{viscous force}} = \frac{\rho V D}{\mu} = \frac{V D}{\nu}$$

What it tells us. Large V, large D or small viscosity make Re large, and the flow turns turbulent. Thick fluids in thin tubes at low speed have small Re and stay laminar.

Flow in a circular pipe	Re
Laminar	below about 2000
Transition	2000 to 4000
Turbulent	above about 4000

Feature	**Laminar**	**Turbulent**
Particle paths	Smooth, parallel layers	Irregular, with eddies and mixing
Velocity profile in a pipe	Parabolic, $u_{\max} = 2V$	Flatter, $u_{\max} \approx 1.2V$
Head loss	Proportional to V	Proportional to $V^{1.75}$ to V^2
Examples	Oil in a thin bearing film, blood in capillaries	Water mains, rivers, air round vehicles

Worked example

Step-by-step method.

1. Convert all data to SI: diameter in m, viscosity in Pa·s or m^2/s.
2. If ν is given use $\text{Re} = VD/\nu$; if μ and ρ are given use $\text{Re} = \rho VD/\mu$.
3. If only Q is given, first find $V = Q/A$.
4. Compare Re with 2000 and 4000 and name the regime.
5. For the limiting velocity, put $\text{Re} = 2000$ and solve for V.

Example 3.0 (warm-up) CORE Water ($\nu = 10^{-6}\ \mathrm{m^2/s}$) flows at $0.05\ \mathrm{m/s}$ in a 20 mm pipe. Find Re and the type of flow.

1. $D = 20\ \mathrm{mm} = 0.02\ \mathrm{m}$
2. $\mathrm{Re} = \dfrac{VD}{\nu} = \dfrac{0.05 \times 0.02}{10^{-6}} = 1000$
3. $1000 < 2000$, so the flow is **laminar**

Example 3.1 CORE Water ($\nu = 10^{-6}\ \mathrm{m^2/s}$) flows in a 50 mm pipe. Find the type of flow at $0.03\ \mathrm{m/s}$ and at $0.1\ \mathrm{m/s}$, and the highest velocity for laminar flow.

1. $V = 0.03\ \mathrm{m/s}$: $\mathrm{Re} = \dfrac{0.03 \times 0.05}{10^{-6}} = 1500$, **laminar**
2. $V = 0.1\ \mathrm{m/s}$: $\mathrm{Re} = \dfrac{0.1 \times 0.05}{10^{-6}} = 5000$, **turbulent**
3. $\mathrm{Re} = 2000$ gives $V = \dfrac{2000 \times 10^{-6}}{0.05} = 0.04\ \mathrm{m/s}$: water flow in ordinary pipes is almost always turbulent

Common mistakes and exam tips.

- Convert mm to m before using Re. A diameter of 50 mm is $0.05\ \mathrm{m}$, not 50.
- Use the pipe **diameter** in Re, not the radius.
- If viscosity is in stokes, multiply by 10^{-4} to get $\mathrm{m^2/s}$. If it is in poise, multiply by 0.1 to get $\mathrm{Pa{\cdot}s}$.
- The limits 2000 and 4000 are for **pipes**. Do not apply them to flow over a plate.

Check your understanding.

1. CORE What does Re measure? *(The ratio of inertia force to viscous force.)*
2. CORE What type of flow has $\mathrm{Re} = 1500$ in a pipe? *(Laminar, since it is below 2000.)*
3. HOTS If the pipe diameter doubles at the same V, what happens to Re? *(It doubles, since $\mathrm{Re} \propto D$.)*

4. Rotational and irrotational flows (L4) CORE

A flow is **rotational** when its fluid particles spin about their own centres as they move, and **irrotational** when they do not. This is about the spin of each particle, not the path it follows: fluid moving in a circle can be irrotational, and fluid moving in straight lines can be rotational.

IN SIMPLE WORDS

Drop a small cork in a river. If the cork turns about its own centre as it floats along, the flow there is rotational. If it keeps pointing the same way, the flow is irrotational, even when it moves in a circle. Spin of the particle is what matters, not the shape of its path. A cork near the river bank, where the water speed changes quickly from one side to the other, usually spins.

4.1 Rotation and vorticity CORE

IN SIMPLE WORDS

Rotation ω_z is the spin rate of a tiny fluid element about the z axis, in rad/s. It arises when the velocity v changes along x, or the velocity u changes along y. Imagine two small sticks, one along x and one along y, floating in the flow. The element's spin is the average turning rate of the two sticks. Vorticity ζ is simply twice the rotation.

The angular velocity of a fluid element about each axis is half the difference of the cross velocity gradients:

$$\omega_z = \frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right), \quad \omega_x = \frac{1}{2}\left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right), \quad \omega_y = \frac{1}{2}\left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right)$$

Derivation from scratch: rotation of a fluid element

Aim. Show that the angular velocity of a fluid element about the z axis is $\omega_z = \frac{1}{2}\left(\partial v/\partial x - \partial u/\partial y\right)$.

Assumptions.

1. The flow is two-dimensional in the x–y plane, with velocity $u\left(x, y\right)$ and $v\left(x, y\right)$.
2. The element is very small, so the velocity changes linearly across it.
3. Anticlockwise rotation is positive.

Step 1. Choose two lines in the element. Take a point O at the corner of a small rectangle. Let OA be a line of length dx along x, and OB a line of length dy along y.

Step 2. Turning of line OA. The velocity in the y direction at O is v. At A, a distance dx along x, it is $v + \frac{\partial v}{\partial x}\,dx$. So A moves upward relative to O with speed $\frac{\partial v}{\partial x}\,dx$. In a small time dt, A moves up by:

$$\frac{\partial v}{\partial x}\,dx\,dt$$

The line has length dx, so for a small angle (angle = displacement / length) it turns anticlockwise through:

$$d\alpha = \frac{\left(\partial v/\partial x\right)\,dx\,dt}{dx} = \frac{\partial v}{\partial x}\,dt$$

Step 3. Turning of line OB. The velocity in the x direction at O is u. At B, a distance dy along y, it is $u + \frac{\partial u}{\partial y}\,dy$. So B moves in the $+x$ direction relative to O by $\frac{\partial u}{\partial y}\,dy\,dt$. For a line pointing along $+y$, a push in the $+x$ direction turns it clockwise. So, anticlockwise:

$$d\beta = -\frac{\left(\partial u/\partial y\right)\,dy\,dt}{dy} = -\frac{\partial u}{\partial y}\,dt$$

Step 4. Angular velocity of each line. Divide the angles by dt.

$$\dot{\alpha} = \frac{\partial v}{\partial x}, \qquad \dot{\beta} = -\frac{\partial u}{\partial y}$$

Step 5. Average. The rotation of the element is the mean of the angular velocities of the two lines.

$$\omega_z = \frac{1}{2}\left(\dot{\alpha} + \dot{\beta}\right) = \frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)$$

Step 6. The other two axes. Repeating with the y–z and z–x planes (cyclic change $x \to y \to z \to x$, $u \to v \to w \to u$) gives ω_x and ω_y.

Result.

$$\omega_z = \frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)$$

$$\omega_x = \frac{1}{2}\left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right), \qquad \omega_y = \frac{1}{2}\left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right)$$

What it tells us. Rotation exists only if v varies with x or u varies with y in a way that does not cancel. Setting $\omega_z = 0$ gives the irrotational condition $\partial v/\partial x = \partial u/\partial y$.

Vorticity is twice the rotation: $\zeta = 2\omega$. The flow is irrotational when the rotation about all three axes is zero everywhere; for a two-dimensional flow in the x–y plane only the rotation about z remains, and the condition reduces to:

$$\frac{\partial v}{\partial x} = \frac{\partial u}{\partial y}$$

4.2 Physical examples HOTS

Flow	Rotational or irrotational	Why
Flow in a boundary layer, pipe flow, any viscous shear flow	Rotational	Velocity gradient du/dy makes particles spin
Forced vortex (fluid in a rotating tank)	Rotational	Fluid turns as a solid body, ω constant
Free vortex (water draining from a sink, away from the core)	Irrotational	$V \times r = \text{constant}$; particles keep their orientation
Flow outside the boundary layer round a body	Practically irrotational	Viscous effects negligible

Worked examples

Step-by-step method.

1. Write the velocity components u and v (and w for 3D).
2. Differentiate: find $\partial v/\partial x$ and $\partial u/\partial y$ separately.
3. Substitute into $\omega_z = \frac{1}{2}\left(\partial v/\partial x - \partial u/\partial y\right)$.
4. If $\omega_z = 0$ everywhere, the flow is irrotational; otherwise it is rotational.
5. State the answer with units (rad/s) and sign (anticlockwise positive). Vorticity is $\zeta = 2\omega$.

Example 4.0 (warm-up) CORE Find the rotation of the flow $u = 4y$, $v = 0$.

1. $\dfrac{\partial v}{\partial x} = 0$; $\dfrac{\partial u}{\partial y} = 4$
2. $\omega_z = \frac{1}{2}(0 - 4) = -2\,\text{rad/s}$: **rotational**, turning clockwise
3. Vorticity $\zeta = 2\omega_z = -4\,\text{s}^{-1}$

Example 4.1 CORE Check whether the flow $u = x^2 - y^2$, $v = -2xy$ is irrotational.

1. $\dfrac{\partial v}{\partial x} = -2y$; $\dfrac{\partial u}{\partial y} = -2y$
2. Rotation about z: $\omega = \frac{1}{2}(-2y - (-2y)) = 0$: **irrotational**

Example 4.2 CORE Find the rotation of the flow $u = 2y$, $v = 3x$.

1. $\dfrac{\partial v}{\partial x} = 3$; $\dfrac{\partial u}{\partial y} = 2$
2. Rotation about z: $\omega = \frac{1}{2}(3 - 2) = 0.5\,\text{rad/s}$: **rotational**; vorticity $\zeta = 1\,\text{s}^{-1}$

Common mistakes and exam tips.

- The order is $\partial v/\partial x$ **minus** $\partial u/\partial y$. Reversing it flips the sign.
- Do not forget the factor $\frac{1}{2}$ for rotation. Vorticity is twice the rotation.
- Differentiate v with respect to x and u with respect to y. Do not differentiate u with respect to x.
- A circular path does not prove rotation. A free vortex moves in circles but is irrotational.

Check your understanding.

1. CORE Write the irrotational condition for 2D flow. *($\partial v/\partial x = \partial u/\partial y$.)*
2. CORE What is the vorticity if $\omega_z = 0.5\,\text{rad/s}$? *($\zeta = 2\omega = 1\,\text{s}^{-1}$.)*
3. HOTS Is water draining from a sink, away from the core, rotational? *(No. It is a free vortex, which is irrotational.)*

5. Compressible and incompressible; subsonic, sonic and supersonic flows (L5) CORE

In **incompressible flow** the density is constant ($\rho = \text{const}$); in **compressible flow** it changes from point to point ($\rho \neq \text{const}$). Liquids are practically incompressible. Gases behave as incompressible at low speed, and the Mach number tells us when that stops being true.

> **IN SIMPLE WORDS**
> Water in a pipe keeps its density almost exactly. Air does not: squeeze it and it packs closer. At low speeds the squeeze is so small that we ignore it. At high speeds, such as a jet aircraft, we cannot. The Mach number tells how fast the flow is compared with the speed of sound. Sound is a small pressure disturbance, so its speed is the speed at which the fluid "gets the news".

5.1 Mach number CORE

> **IN SIMPLE WORDS**
> $M = V/c$ is the flow speed divided by the speed of sound. Here V is the speed of the gas or the body, and c is the speed of sound in that gas. At $M = 0.5$ the flow moves at half the speed of sound. Below $M = 1$ a sound wave can outrun the flow, so the flow is subsonic. Above $M = 1$ the flow outruns its own sound, so it is supersonic. Sound speed in a gas depends on temperature, which must be in kelvin.

$$M = \frac{V}{c}, \qquad c = \sqrt{\frac{K}{\rho}} \ \text{(any fluid)}, \qquad c = \sqrt{kRT} \ \text{(ideal gas)}$$

For air ($k = 1.4$, $R = 287\ \text{J/kg}\cdot\text{K}$) at 15 °C, $c \approx 340\ \text{m/s}$. Density changes are below about 5% for $M < 0.3$, so such gas flows are treated as incompressible.

Regime	Mach number	Behaviour	Example
Incompressible	$M < 0.3$	Density practically constant	Fans, ventilation ducts, cars
Subsonic	$M < 1$	Pressure waves travel ahead of the body and warn the flow	Passenger aircraft at cruise ($M \approx 0.8$)
Transonic	about 0.8 to 1.2	Mixed subsonic and supersonic regions	Transport aircraft near $M = 1$
Sonic	$M = 1$	Waves pile up at the body; flow chokes in a nozzle throat	Throat of a choked nozzle
Supersonic	$M > 1$	Body outruns its waves; Mach cone and shock waves form	Fighter aircraft, rifle bullets
Hypersonic	$M > 5$	Strong shocks, high temperatures	Re-entry vehicles

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

In supersonic flow the disturbances are confined inside a **Mach cone** whose half-angle α is given by:

$$\sin \alpha = \frac{c}{V} = \frac{1}{M}$$

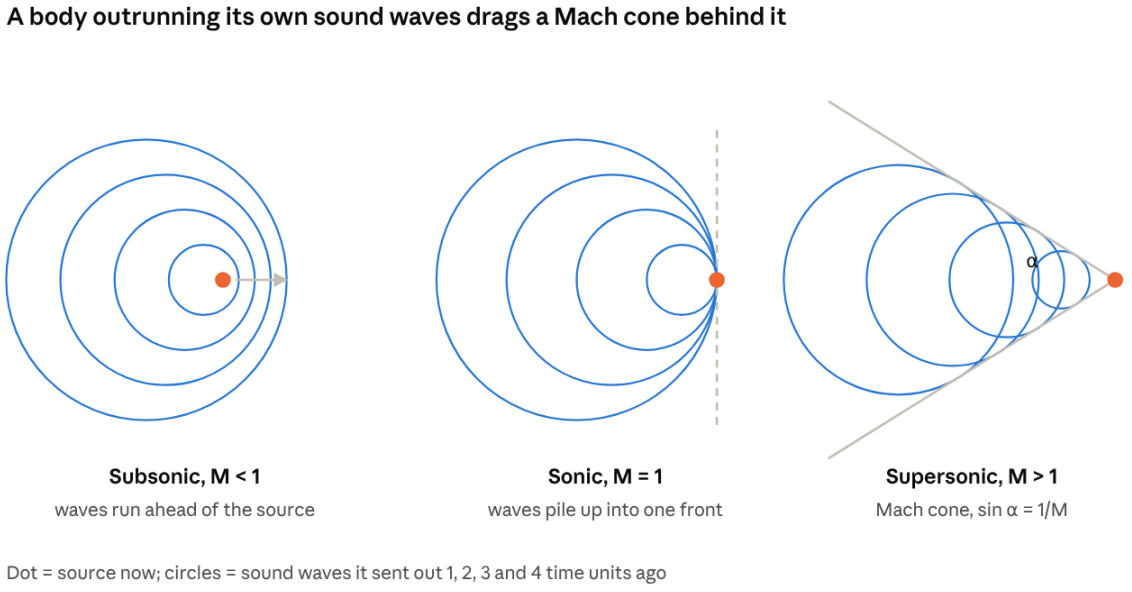


Figure 3. Sound waves from a moving source at subsonic, sonic and supersonic speed

Worked examples

Step-by-step method.

1. Convert the speed to m/s (km/h divided by 3.6).
2. Convert the temperature to kelvin (add 273 to the Celsius value).
3. Find the speed of sound from $c = \sqrt{kRT}$.
4. Find $M = V/c$.
5. Classify: $M < 0.3$ incompressible, $M < 1$ subsonic, $M = 1$ sonic, $M > 1$ supersonic.

Example 5.0 (warm-up) CORE A gas stream moves at $170\ \mathrm{m/s}$ where the speed of sound is $340\ \mathrm{m/s}$. Find M and classify the flow.

1. $M = \dfrac{V}{c} = \dfrac{170}{340} = 0.5$
2. $0.3 < 0.5 < 1$, so the flow is **subsonic** and compressibility cannot be ignored

Example 5.1 CORE An aircraft flies at 900 km/h where the air temperature is $-50\ ^\circ\mathrm{C}$. Find the Mach number. ($k = 1.4$, $R = 287\ \mathrm{J/kg \cdot K}$)

1. $c = \sqrt{1.4 \times 287 \times 223} = 299.3\ \mathrm{m/s}$; $V = \dfrac{900}{3.6} = 250\ \mathrm{m/s}$
2. $M = \dfrac{250}{299.3} = 0.835$: **subsonic**

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

Example 5.2 ADVANCED A bullet travels at $600\ \mathrm{m/s}$ through air at 15 °C. Find the Mach number and the Mach angle.

1. $c = \sqrt{1.4 \times 287 \times 288} = 340.2\ \mathrm{m/s}$; $M = \dfrac{600}{340.2} = 1.76$: **supersonic**
2. $\alpha = \sin^{-1}\left(\dfrac{1}{1.76}\right) = 34.5^\circ$

Common mistakes and exam tips.

- Use absolute temperature in kelvin in $c = \sqrt{kRT}$. Using °C gives a wrong c.
- Convert km/h to m/s by dividing by 3.6, not multiplying.
- Do not confuse the two limits: $M < 0.3$ means incompressible, while $M < 1$ means subsonic. A flow at $M = 0.6$ is subsonic but not incompressible.
- The speed of sound is not a fixed number. It changes with temperature (about $340\ \mathrm{m/s}$ only near 15 °C).

Check your understanding.

1. CORE Below what Mach number can a gas flow be treated as incompressible? *($M < 0.3$.)*
2. CORE What is the flow called if $M = 1$? *(Sonic.)*
3. HOTS Why does an aircraft at $M = 0.85$ still get "warning" ahead of it? *(Sound waves outrun it, as the flow is subsonic.)*

6. Sub-critical, critical and supercritical flows (L6) CORE

This classification applies to **open-channel flow** (rivers, canals, spillways), where the liquid has a free surface. It compares the flow velocity V with the speed $\sqrt{gy}$ of a small surface wave in a channel of depth y, through the Froude number:

$$\mathrm{Fr} = \frac{V}{\sqrt{gy}} = \sqrt{\frac{\text{inertia force}}{\text{gravity force}}}$$

IN SIMPLE WORDS

Throw a pebble into a still pond and ripples spread out at a certain speed. In a canal this ripple speed is $\sqrt{gy}$, where y is the depth. If the water flows slower than the ripples, a ripple can swim upstream and warn the water ahead: sub-critical. If the water flows faster, ripples are swept downstream: supercritical. The Froude number Fr is just flow speed divided by ripple speed.

Flow	Froude number	Behaviour	Example
Sub-critical (tranquil, streaming)	$\mathrm{Fr} < 1$	Deep, slow; surface waves can travel upstream, so downstream conditions affect the flow	Rivers in plains, flow upstream of a weir
Critical	$\mathrm{Fr} = 1$	Wave speed equals flow speed; specific energy is minimum	Crest of a broad-crested weir, control sections
Supercritical (rapid, shooting)	$\mathrm{Fr} > 1$	Shallow, fast; waves are swept downstream, so only upstream conditions matter	Spillway chutes, flow under a sluice gate

The Froude number plays the same role in open channels that the Mach number plays in gas flow: both compare the flow speed with the speed of a small disturbance. A supercritical stream that is forced to slow down does so abruptly in a **hydraulic jump**, much as a supersonic flow slows through a shock wave.

Derivation from scratch: critical flow occurs at $\mathrm{Fr} = 1$

Aim. Show that the flow with the minimum specific energy has $V = \sqrt{gy}$, so that $\mathrm{Fr} = 1$.

Assumptions.

1. The channel is rectangular with width b and a horizontal bed.
2. The discharge Q is fixed and the flow is steady and uniform across the section.
3. Specific energy E is the energy per unit weight measured above the channel bed.

Step 1. Specific energy. It is the depth plus the velocity head.

$$E = y + \frac{V^2}{2g}$$

Step 2. Write V using the discharge per unit width. Let $q = Q/b$. Then $V = Q/(by) = q/y$.

$$E = y + \frac{q^2}{2g\,y^2}$$

Step 3. Find the depth for minimum E. Differentiate with respect to y at constant q and set the derivative to zero.

$$\frac{dE}{dy} = 1 - \frac{q^2}{g\,y^3} = 0 \quad \Rightarrow \quad q^2 = g\,y_c^3$$

Step 4. Check that this is a minimum. The second derivative is positive.

$$\frac{d^2E}{dy^2} = \frac{3q^2}{g\,y^4} > 0$$

Step 5. Put $q = V_c y_c$ and simplify.

$$V_c^2\,y_c^2 = g\,y_c^3 \quad \Rightarrow \quad V_c^2 = g\,y_c$$

Result.

$$V_c = \sqrt{g\,y_c} \quad \Rightarrow \quad \mathrm{Fr} = \frac{V_c}{\sqrt{g\,y_c}} = 1$$

What it tells us. At the minimum specific energy the flow speed equals the speed of a small surface wave. For $y > y_c$ the flow is slower and deeper ($\mathrm{Fr} < 1$); for $y < y_c$ it is faster and shallower ($\mathrm{Fr} > 1$).

Worked example

Step-by-step method.

1. Write down the velocity V and the depth y in metres.
2. Compute the wave speed $\sqrt{gy}$ with $g = 9.81\ \mathrm{m/s^2}$.
3. Compute $\mathrm{Fr} = V/\sqrt{gy}$.
4. Compare with 1: below is sub-critical, equal is critical, above is supercritical.
5. For the critical velocity at a given depth, use $V_c = \sqrt{gy}$.

Example 6.0 (warm-up) CORE Water flows at $2\ \mathrm{m/s}$ in a channel $1\ \mathrm{m}$ deep. Find Fr and classify the flow.

1. $\sqrt{gy} = \sqrt{9.81 \times 1} = 3.13\ \mathrm{m/s}$
2. $\mathrm{Fr} = \dfrac{2}{3.13} = 0.64$
3. $0.64 < 1$, so the flow is **sub-critical**

Example 6.1 CORE Classify the flow (a) at $3\ \mathrm{m/s}$ in a channel 0.5 m deep and (b) at $1.5\ \mathrm{m/s}$ in a channel 2 m deep. Find the critical velocity for case (a).

1. (a) $\mathrm{Fr} = \dfrac{3}{\sqrt{9.81 \times 0.5}} = 1.35$: **supercritical**
2. (b) $\mathrm{Fr} = \dfrac{1.5}{\sqrt{9.81 \times 2}} = 0.34$: **sub-critical**
3. Critical velocity at 0.5 m depth: $V = \sqrt{9.81 \times 0.5} = 2.21\ \mathrm{m/s}$

Common mistakes and exam tips.

- Use the **depth** y of the flow, not the width or the channel height, in $\sqrt{gy}$.
- Fr is for open channels with a free surface. Do not use the pipe Re limits here.
- Sub-critical means $\mathrm{Fr} < 1$ (slow and deep). Students often reverse sub- and supercritical.
- The Froude number has no unit. If you get a unit, a quantity was left unconverted.

Check your understanding.

1. CORE What is Fr at critical flow? *($\mathrm{Fr} = 1$.)*
2. CORE Can a surface wave travel upstream in supercritical flow? *(No, the flow carries it downstream.)*
3. HOTS What happens when a supercritical stream is forced to slow down? *(It does so abruptly in a hydraulic jump.)*

7. One-, two- and three-dimensional flows (L6) CORE

The dimension of a flow is the number of space coordinates needed to describe its velocity field. Real flows are three-dimensional; engineers simplify them to one or two dimensions whenever the variation in the other directions is small.

> **IN SIMPLE WORDS**
>
> Counting dimensions means counting how many directions you need to describe the speed. In a long straight pipe, the speed changes mainly along the pipe, so one number at each section is enough. For water flowing over a very wide weir, the picture is the same in every slice, so two coordinates are enough. Fewer dimensions give much easier equations. Engineers choose the simplest model that still gives a good answer.

Flow	Velocity depends on	Written as	Example
One-dimensional	One space coordinate (and time)	$u = f(x)$, $v = w = 0$	Pipe flow described by the mean velocity at each section
Two-dimensional	Two space coordinates	$u = f_1(x, y)$, $v = f_2(x, y)$, $w = 0$	Flow over a long wing or weir, between parallel plates, round a long cylinder
Three-dimensional	Three space coordinates	u, v, w all functions of (x, y, z)	Flow round a car, in a river bend, through a pump impeller

One-dimensional analysis is the workhorse of this course. In a pipe the velocity actually varies across the section (zero at the wall, maximum at the centre), but replacing it by the mean velocity $V = Q/A$ at each section turns the problem into one variable along the pipe. Bernoulli's equation as used for flow meters, and the 1D continuity equation in Section 9, both rest on this simplification.

> **Common mistakes and exam tips.**
>
> - The dimension of a flow counts the space coordinates the velocity depends on, not the number of velocity components.
> - Pipe flow is treated as 1D by using the **mean** velocity $V = Q/A$ at each section; the true profile is not uniform.
> - Be ready to give one example each of 1D, 2D and 3D flow from the table.
> - Real flows are always 3D; 1D and 2D are modelling choices, so say "assumed" in your answer.

Check your understanding.

1. CORE How many coordinates does a 2D flow depend on? *(Two, such as x and y.)*
2. CORE Give one example of 3D flow. *(Flow round a car, a river bend or a pump impeller.)*

3. HOTS Why can pipe flow be treated as 1D? *(Using the mean velocity at each section leaves only the distance along the pipe as a variable.)*

8. Streamlines, pathlines, streaklines and stream tubes (L7)

CORE

Four kinds of line are used to picture a flow. In steady flow the first three coincide; in unsteady flow they differ.

> **IN SIMPLE WORDS**
> These lines are different ways of drawing the same flow. A **streamline** is a snapshot: arrows that show the direction of motion everywhere at one instant. A **pathline** is the track of one particle, like a long-exposure photo of a firefly. A **streakline** is what smoke from a chimney shows: all particles that passed one point. In steady flow all three look the same.

Line	Definition	How it is seen
Streamline	An imaginary line drawn so that the velocity at every point on it is tangent to it, at one instant	Short exposure photograph of many tracer particles
Pathline	The actual path traced by one fluid particle over a period of time	Long exposure photograph of one glowing particle
Streakline	The line joining all particles that have passed through one fixed point	Dye or smoke released continuously from one point
Stream tube	A tube whose surface is made of streamlines through a small closed curve	Behaves like a pipe: no fluid crosses its wall

Properties of streamlines:

- Two streamlines cannot cross, since a point would then have two velocities (except at a stagnation point, where $V = 0$).
- No fluid crosses a streamline, so the discharge between two streamlines is constant.
- Streamlines crowd together where the velocity is high and spread out where it is low.

8.1 Equation of a streamline CORE

> **IN SIMPLE WORDS**
> On a streamline the velocity arrow points along the line. So the slope of the line at each point equals the ratio of the velocity components. In 2D this says $dy/dx = v/u$. Rearranged, it is $dx/u = dy/v$. Integrating this gives the streamline pattern, with one constant C for each streamline.

Since the velocity is tangent to the streamline, the small element ds of a streamline is parallel to V, so its components are in the same ratio as u, v, w:

$$\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w}$$

Derivation from scratch: equation of a streamline

Aim. Show that along a streamline $dx/u = dy/v = dz/w$.

Assumptions.

1. Consider one instant of time, so the velocity field is fixed.
2. A streamline is a curve whose tangent at every point is along the local velocity $\vec{V} = u\hat{i} + v\hat{j} + w\hat{k}$.

Step 1. Element of the streamline. Take a small step along the curve:

$$d\vec{s} = dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}$$

Step 2. Parallel vectors. The step $d\vec{s}$ lies along the tangent, which is along $\vec{V}$. Two parallel vectors differ only by a scale factor λ:

$$dx = \lambda u, \qquad dy = \lambda v, \qquad dz = \lambda w$$

Step 3. Eliminate λ. Divide each equation by its velocity component.

$$\frac{dx}{u} = \lambda, \qquad \frac{dy}{v} = \lambda, \qquad \frac{dz}{w} = \lambda$$

Result.

$$\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w}$$

What it tells us. In two dimensions this is the slope statement $dy/dx = v/u$. Integrating gives one constant for each streamline, and different values of the constant give the different streamlines of the pattern.

Worked example

Step-by-step method.

1. Write the streamline equation $dx/u = dy/v$ with the given u and v.
2. Separate the variables: all x terms on one side, all y terms on the other.
3. Integrate both sides, and add one constant of integration.
4. Combine logarithms (write the constant as $\ln C$) to get the final relation.
5. Name the pattern (straight line, hyperbola, circle) and note any stagnation point where $u = v = 0$.

Example 8.0 (warm-up) CORE Find the streamlines for the uniform flow $u = 3\,\mathrm{m/s}$, $v = 6\,\mathrm{m/s}$.

1. $\dfrac{dx}{3} = \dfrac{dy}{6}$, so $dy = 2\,dx$
2. Integrating: $y = 2x + C$
3. The streamlines are parallel straight lines with slope 2, along the direction of the velocity

Example 8.1 HOTS Find the equation of the streamlines for the two-dimensional flow $u = 2x$, $v = -2y$.

1. $\dfrac{dx}{2x} = \dfrac{dy}{-2y}$, so $\dfrac{dx}{x} = -\dfrac{dy}{y}$
2. Integrating: $\ln x = -\ln y + \ln C$, so $xy = C$
3. The streamlines are rectangular hyperbolas: this is the flow into a 90° corner, with a stagnation point at the origin

Common mistakes and exam tips.

- Streamlines, pathlines and streaklines coincide only in **steady** flow.
- Two streamlines never cross, except at a stagnation point where $V = 0$.
- Integrate $dx/u = dy/v$ by separating variables; do not forget the constant C (or $\ln C$).
- A stream tube has no flow through its wall, so the discharge along it is constant.

Check your understanding.

1. CORE Write the equation of a streamline. *($dx/u = dy/v = dz/w$.)*
2. CORE When do streamline, pathline and streakline coincide? *(In steady flow.)*
3. HOTS Why can two streamlines not cross? *(The point of crossing would have two different velocity directions.)*

9. Continuity equation (L8) CORE

The continuity equation is conservation of mass for a flowing fluid: the net rate at which mass flows out of a fixed volume equals the rate at which the mass inside it decreases. For steady flow, whatever mass enters must leave.

IN SIMPLE WORDS

Mass cannot be created or destroyed. If water enters a pipe at one end, the same mass must leave at the other end, unless the pipe is filling up. Continuity just writes this “no loss, no gain” rule as an equation. It lets us find an unknown velocity from a known one, and it tells us whether a given velocity field could really exist.

9.1 Three-dimensional continuity equation (Cartesian) CORE

IN SIMPLE WORDS

Imagine a tiny box in the flow. Fluid enters through some faces and leaves through others. If more mass leaves than enters, the density inside must fall. The equation adds up the in and out flow through all six faces. Here ρu is the mass flowing per second through each unit area, in the x direction. For water (ρ constant) the rule becomes: the velocity gradients $\partial u/\partial x$, $\partial v/\partial y$, $\partial w/\partial z$ must add to zero.

Take a small fixed rectangular element with sides dx, dy, dz, through which fluid of density ρ flows with velocity components u, v, w.

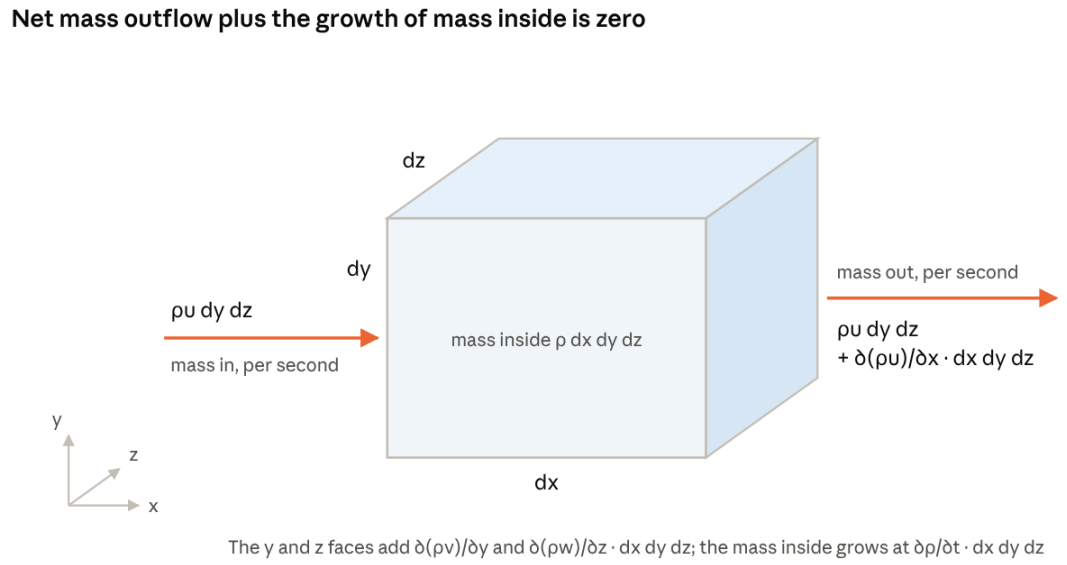


Figure 4. Fluid element with the mass flow through its x-faces

Derivation from scratch: continuity equation in Cartesian coordinates

Aim. Prove that for any flow, steady or unsteady, compressible or incompressible,
$\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u)}{\partial x} + \frac{\partial (\rho v)}{\partial y} + \frac{\partial (\rho w)}{\partial z} = 0.$

Assumptions.

1. Mass is conserved (no source or sink inside the element).
2. The element is a fixed rectangular box with sides dx, dy, dz, so its volume $dx\,dy\,dz$ is constant.
3. The fluid is a continuum with density $\rho (x, y, z, t)$ and velocity components u, v, w .
4. The element is so small that the change of a quantity across it is given by the first term of its Taylor series.

Step 1. Mass flow through the left face (the x-direction). The mass flux (mass per second per unit area) in the x direction is ρu. The left face has area $dy\,dz$ and lies at position x.

$$\text{Mass in per second} = \rho u\,dy\,dz$$

Step 2. Mass flow through the right face. The right face lies at $x + dx$. The mass flux there is the value at x plus its rate of change times dx.

$$\text{Mass out per second} = \left[\rho u + \frac{\partial\,(\,\rho u\,)}{\partial x}\,dx \right] dy\,dz$$

Step 3. Net outflow in the x-direction. Subtract.

$$\text{Net outflow}_x = \left[\rho u + \frac{\partial\,(\,\rho u\,)}{\partial x}\,dx \right] dy\,dz - \rho u\,dy\,dz = \frac{\partial\,(\,\rho u\,)}{\partial x}\,dx\,dy\,dz$$

Step 4. Net outflow in the y- and z-directions. The same argument for the front and back faces (area $dx\,dz$, flux ρv) and for the top and bottom faces (area $dx\,dy$, flux ρw) gives:

$$\text{Net outflow}_y = \frac{\partial\,(\,\rho v\,)}{\partial y}\,dx\,dy\,dz, \qquad \text{Net outflow}_z = \frac{\partial\,(\,\rho w\,)}{\partial z}\,dx\,dy\,dz$$

Step 5. Total net outflow.

$$\text{Total net outflow} = \left[\frac{\partial\,(\,\rho u\,)}{\partial x} + \frac{\partial\,(\,\rho v\,)}{\partial y} + \frac{\partial\,(\,\rho w\,)}{\partial z} \right] dx\,dy\,dz$$

Step 6. Rate of increase of mass inside. The mass inside is $\rho\,dx\,dy\,dz$. The volume is fixed, so only ρ can change.

$$\text{Rate of increase} = \frac{\partial \rho}{\partial t}\,dx\,dy\,dz$$

Step 7. Mass balance. Conservation of mass says: net outflow = rate of decrease of mass inside, that is, net outflow + rate of increase = 0.

$$\frac{\partial \rho}{\partial t}\,dx\,dy\,dz + \left[\frac{\partial\,(\,\rho u\,)}{\partial x} + \frac{\partial\,(\,\rho v\,)}{\partial y} + \frac{\partial\,(\,\rho w\,)}{\partial z} \right] dx\,dy\,dz = 0$$

Step 8. Divide by $dx\,dy\,dz$.

Result.

$$\frac{\partial \rho}{\partial t} + \frac{\partial\,(\,\rho u\,)}{\partial x} + \frac{\partial\,(\,\rho v\,)}{\partial y} + \frac{\partial\,(\,\rho w\,)}{\partial z} = 0$$

Step 9. Special case: incompressible flow. For constant ρ, $\partial\rho/\partial t = 0$ and ρ comes out of each derivative.

$$\rho \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) = 0 \quad \Rightarrow \quad \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

Step 10. Special case: steady flow. Put $\partial\rho/\partial t = 0$ in the result to get the steady compressible form in the table below. For two-dimensional incompressible flow, $w = 0$ and nothing depends on z, so $\partial u/\partial x + \partial v/\partial y = 0$.

What it tells us. If the net outflow is positive, the density inside falls. In steady flow every kilogram that enters must leave. For a liquid the sum of the three velocity gradients must be zero: if the fluid stretches in one direction it must squeeze in another.

The boxed result is the general continuity equation, valid for steady or unsteady, compressible or incompressible flow. Its special cases:

Case	Continuity equation
Steady, compressible	$\partial(\rho u)/\partial x + \partial(\rho v)/\partial y + \partial(\rho w)/\partial z = 0$
Incompressible (steady or unsteady)	$\partial u/\partial x + \partial v/\partial y + \partial w/\partial z = 0$
Two-dimensional, incompressible	$\partial u/\partial x + \partial v/\partial y = 0$

In vector form: $\partial\rho/\partial t + \nabla\cdot(\rho\mathbf{V}) = 0$, and for incompressible flow $\nabla\cdot\mathbf{V} = 0$. A velocity field that does not satisfy the incompressible form cannot describe a real liquid flow.

9.2 One-dimensional continuity equation CORE

IN SIMPLE WORDS

Think of a river that narrows. The same amount of water must pass every second, so the water speeds up in the narrow part. In a pipe, discharge $Q = AV$ is the volume per second. If no water leaks out, Q is the same at every section. For gases the density can change, so we track the mass per second, $\dot{m} = \rho AV$, instead.

For steady flow through a stream tube or pipe between sections 1 and 2, no mass crosses the wall, so the mass flow rate is the same at every section.

Derivation from scratch: one-dimensional continuity equation

Aim. Show that $\rho_1 A_1 V_1 = \rho_2 A_2 V_2$ and, for incompressible flow, $A_1 V_1 = A_2 V_2$.

Assumptions.

1. The flow is steady, so the mass inside the stream tube between sections 1 and 2 does not change.
2. No fluid crosses the wall of the stream tube or pipe.
3. At each section the density ρ and the mean velocity V are uniform over the area A.

Step 1. Mass passing a section. In a short time dt the fluid at a section moves a distance Vdt. The fluid that crosses is a column of area A and length Vdt. Its volume is $AVdt$ and its mass is $\rho AVdt$.

$$\dot{m} = \frac{\rho AVdt}{dt} = \rho AV$$

Step 2. Mass balance. In steady flow, mass entering at section 1 equals mass leaving at section 2.

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2$$

Step 3. Incompressible flow. If $\rho_1 = \rho_2 = \rho$, cancel it from both sides.

Result.

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2 \qquad \text{incompressible: } Q = A_1 V_1 = A_2 V_2$$

What it tells us. In a pipe that narrows, the velocity must rise in the inverse ratio of the areas. Since $A \propto D^2$, halving the diameter quadruples the velocity.

For a pipe that branches, the flow into the junction equals the total flow out: $Q_1 = Q_2 + Q_3$.

Worked examples

Step-by-step method.

1. For a velocity field: differentiate u with respect to x, v with respect to y (and w with respect to z).
2. Add the derivatives. If the sum is zero, continuity is satisfied and the incompressible flow is possible.
3. To find a missing component w: integrate $\partial w/\partial z = -\left(\partial u/\partial x + \partial v/\partial y\right)$ with respect to z, and fix the constant of integration from the given condition.
4. For a pipe: convert diameters to metres and find $A = \frac{\pi}{4}D^2$.
5. Use $Q = AV$ and $A_1V_1 = A_2V_2$. At a junction, flow in equals total flow out.
6. For a compressible gas use $\rho_1 A_1 V_1 = \rho_2 A_2 V_2$.

Example 9.0 (warm-up) CORE Water flows at $0.5\,\text{m/s}$ in a 100 mm pipe that narrows to 50 mm. Find the discharge and the velocity in the 50 mm pipe.

1. $Q = A_1V_1 = \dfrac{\pi}{4} \times 0.1^2 \times 0.5 = 0.00393\,\text{m}^3/\text{s}$
2. $V_2 = V_1\left(\dfrac{D_1}{D_2}\right)^2 = 0.5 \times \left(\dfrac{0.1}{0.05}\right)^2 = 2\,\text{m/s}$

Example 9.1 CORE Check whether $u = x^2 + y^2$, $v = -2xy$ satisfies the continuity equation for incompressible flow.

$\dfrac{\partial u}{\partial x} + \dfrac{\partial v}{\partial y} = 2x - 2x = 0$, so **the flow is possible**.

Example 9.2 HOTS For $u = x^2 + 2xz$ and $v = y^2 + 2yz$, find w for incompressible flow, given $w = 0$ at $z = 0$.

1. $\dfrac{\partial w}{\partial z} = -\left(\dfrac{\partial u}{\partial x} + \dfrac{\partial v}{\partial y}\right) = -\left(2x + 2z + 2y + 2z\right) = -2\left(x + y\right) - 4z$
2. Integrating: $w = -2\left(x + y\right)z - 2z^2$ (the function of x and y is zero, since $w = 0$ at $z = 0$)

Example 9.3 CORE Water flows at $1.2\,\text{m/s}$ in a 300 mm pipe that reduces to 200 mm. Find the discharge and the velocity in the smaller pipe.

1. $Q = \dfrac{\pi}{4} \times 0.3^2 \times 1.2 = 0.0848\,\text{m}^3/\text{s}$
2. $V_2 = V_1\left(\dfrac{D_1}{D_2}\right)^2 = 1.2 \times \left(\dfrac{0.3}{0.2}\right)^2 = 1.2 \times 2.25 = 2.7\,\text{m/s}$

Example 9.4 HOTS A 450 mm pipe carrying water at $1.5\,\mathrm{m/s}$ divides into a 300 mm branch, where the velocity is $2\,\mathrm{m/s}$, and a 200 mm branch. Find the velocity in the 200 mm branch.

1. $Q_1 = \frac{\pi}{4} \times 0.45^2 \times 1.5 = 0.2386\,\mathrm{m^3/s}$; $Q_2 = \frac{\pi}{4} \times 0.3^2 \times 2 = 0.1414\,\mathrm{m^3/s}$
2. $Q_3 = 0.2386 - 0.1414 = 0.0972\,\mathrm{m^3/s}$
3. $V_3 = \dfrac{0.0972}{(\pi/4) \times 0.2^2} = 3.09\,\mathrm{m/s}$

Example 9.5 (compressible) CORE Air flows steadily through a duct. At section 1, $A = 0.05\,\mathrm{m^2}$, $\rho = 1.2\,\mathrm{kg/m^3}$ and $V = 100\,\mathrm{m/s}$; at section 2, $A = 0.03\,\mathrm{m^2}$ and $\rho = 1.0\,\mathrm{kg/m^3}$. Find the mass flow rate and the velocity at section 2.

1. $\dot{m} = 1.2 \times 0.05 \times 100 = 6\,\mathrm{kg/s}$
2. $V_2 = \dfrac{6}{1.0 \times 0.03} = 200\,\mathrm{m/s}$

Common mistakes and exam tips.

- Velocity depends on the **square** of the diameter ratio: $V_2 = V_1\,(D_1/D_2)^2$, not (D_1/D_2).
- Convert mm to m before finding the area. The area of a 300 mm pipe is $\frac{\pi}{4} \times 0.3^2$, not $\frac{\pi}{4} \times 300^2$.
- At a junction, add **discharges** (not velocities): $Q_1 = Q_2 + Q_3$.
- When integrating to find w, the "constant" is a function of x and y; use the given boundary condition to find it.
- Use the incompressible form only for liquids or slow gas flow; use ρAV for fast gas flow.

Check your understanding.

1. CORE Write the continuity equation for steady incompressible 3D flow. *($\partial u/\partial x + \partial v/\partial y + \partial w/\partial z = 0$.)*
2. CORE A pipe's diameter is halved. What happens to the velocity? *(It becomes four times larger, since $V \propto 1/D^2$.)*
3. HOTS Why does a velocity field that fails the incompressible continuity test not represent a real liquid flow? *(The liquid would have to be created or destroyed somewhere.)*

10. Formula sheet and practice questions CORE

10.1 Formula sheet CORE

Topic	Result
Knudsen number	$\mathrm{Kn} = \lambda / L$; continuum < 0.01, free molecular > 10
Steady flow	$\partial V / \partial t = 0$ at a point
Uniform flow	$\partial V / \partial s = 0$ at an instant
Reynolds number	$\mathrm{Re} = \rho V D / \mu = V D / \nu$; pipe laminar < 2000, turbulent > 4000
Rotation (2D)	$\omega = \frac{1}{2} \left(\partial v / \partial x - \partial u / \partial y \right)$; vorticity $\zeta = 2\omega$; irrotational if $\partial v / \partial x = \partial u / \partial y$
Mach number	$M = V / c$; $c = \sqrt{K / \rho} = \sqrt{kRT}$; incompressible below $M = 0.3$
Mach angle	$\sin \alpha = 1 / M$
Froude number	$\mathrm{Fr} = V / \sqrt{gy}$; sub-critical < 1, critical $= 1$, supercritical > 1
Streamline	$dx / u = dy / v = dz / w$
Continuity, general	$\partial \rho / \partial t + \partial \left(\rho u \right) / \partial x + \partial \left(\rho v \right) / \partial y + \partial \left(\rho w \right) / \partial z = 0$
Continuity, incompressible	$\partial u / \partial x + \partial v / \partial y + \partial w / \partial z = 0$
Continuity, 1D	$\rho_1 A_1 V_1 = \rho_2 A_2 V_2$; incompressible $Q = A_1 V_1 = A_2 V_2$

10.2 Short-answer questions (2 marks) CORE

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

1. ADVANCED Distinguish between the Lagrangian and Eulerian descriptions of fluid motion.

2. CORE What is the Knudsen number? Distinguish continuum flow from free molecular flow.
3. CORE Define steady and uniform flow, with one example of each combination.
4. CORE What is the physical meaning of the Reynolds number?
5. HOTS Distinguish between rotational and irrotational flow. Is a free vortex rotational?
6. CORE Define the Mach number. When can a gas flow be treated as incompressible?

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

7. ADVANCED What is a Mach cone? Write the expression for the Mach angle.

8. CORE Distinguish sub-critical, critical and supercritical flows.
9. HOTS Why can pipe flow be treated as one-dimensional?
10. CORE Define streamline, pathline and streakline. When do they coincide?
11. HOTS Why can two streamlines not cross each other?
12. CORE Write the continuity equation for steady incompressible 3D flow.

10.3 Long-answer and numerical questions (10 marks) CORE

1. CORE Classify fluid flows into their types, with a definition and an example of each.
2. CORE Describe Reynolds' experiment and explain laminar, transition and turbulent flow.
3. CORE Derive the continuity equation in Cartesian coordinates for three-dimensional, unsteady, compressible flow, and reduce it for steady incompressible flow.
4. CORE Explain subsonic, sonic and supersonic flow with sketches of the wave patterns from a moving source.
5. CORE Derive the equation of a streamline and explain the properties of streamlines and stream tubes.
6. CORE Oil ($\nu = 0.5$ stoke) flows at $0.8\ \mathrm{m/s}$ in a 100 mm pipe. Find the Reynolds number and the type of flow. *(Ans: 1600, laminar)*
7. HOTS Check whether the flow $u = 2x$, $v = -2y$ is (a) possible and (b) irrotational. *(Ans: possible; irrotational)*
8. CORE For the flow $u = 3y^2$, $v = 2x$, find the rotation at the point $(1, 1)$. *(Ans: $-2\ \mathrm{rad/s}$)*

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

9. ADVANCED An aircraft flies at 1500 km/h in air at 20 °C. Find the Mach number and the Mach angle. *(Ans: 1.21, 55.4°)*

10. CORE Water flows at $4\ \mathrm{m/s}$ in a channel 1.2 m deep. Classify the flow. *(Ans: $\mathrm{Fr} = 1.17$, supercritical)*
11. CORE Find the equation of the streamlines for $u = y$, $v = x$. *(Ans: $x^2 - y^2 = C$)*
12. HOTS For incompressible flow with $u = 2xy$ and $v = a\left(x^2 - y^2\right)$, find a. *(Ans: $a = 1$)*
13. HOTS Find w for incompressible flow with $u = x^2 + 2xz$ and $v = y^2 + 2yz$, given $w = 0$ at $z = 0$. *(Ans: $w = -2(x + y)z - 2z^2$)*
14. CORE Water at $0.75\ \mathrm{m/s}$ in a 200 mm pipe passes into a 100 mm pipe. Find the discharge and the velocity in the smaller pipe. *(Ans: $0.0236\ \mathrm{m^3/s}$, $3\ \mathrm{m/s}$)*

15. CORE Gas enters a duct of area $0.1\ \mathrm{m}^2$ at $50\ \mathrm{m/s}$ with density $2.0\ \mathrm{kg/m}^3$ and leaves through $0.08\ \mathrm{m}^2$ at $80\ \mathrm{m/s}$. Find the mass flow rate and the exit density. *(Ans:* $10\ \mathrm{kg/s}$*,* $1.56\ \mathrm{kg/m}^3$*)*